

12

STATISTICS AND PROBABILITY

PROBABILITY

Probability, the study of chance, was first used in France to solve problems in gambling in the 1600s. In 1654, the writer and gambler, the Chevalier de Méré, asked mathematician Blaise Pascal many questions involving gambling, including how many rolls of a pair of dice were required to have a better than 50% chance of rolling a double 6. Pascal wrote to another mathematician, Pierre de Fermat, about the problems, and the correspondence between them is regarded to be the beginning of probability theory.

Probability is now used widely in many areas, including determining the survival rate of newborn turtles, predicting the sales of a new version of smartphone and forecasting the weather for a week ahead.



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Chapter outline

	Working mathematically				
12.01 Probability	U	F	PS	R	C
12.02 Observed probability	U	F		R	C
12.03 Venn diagrams*	U	F		R	C
12.04 Two-way tables*	U	F		R	C
12.05 Tree diagrams	U	F	PS	R	C

***PATH**

Wordbank

complementary event The 'opposite' event, for example the complementary event to selecting an Ace from a deck of cards is selecting a card that is not an Ace

relative frequency The frequency of an event over repeated trials as a fraction of the total number of trials, used to calculate **observed probability**

tree diagram A diagram that uses branches to list all the possible outcomes in a multi-stage chance experiment

trial One go or run of a repeated probability experiment, for example, one roll of a die

two-stage experiment A chance experiment with 2 steps or stages, such as rolling a pair of dice

two-way table A way of grouping items into 2 overlapping categories, such as gender and the ability to drive a car

Venn diagram A diagram of circles (usually overlapping) for grouping items into categories

Quiz
Wordbank 12

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (11):

12.01 Probability • Complementary events
12.02 Observed and expected frequencies • Experimental probability
12.03 Venn diagrams 2 • Venn diagrams 1
12.04 Two-way tables 2
12.05 Tables and tree diagrams • Tree diagrams 1 • Tree diagrams 2 • Tree diagrams 3

***Twig* videos (6):**

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***PhET* interactive (1):**

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Quizzes (5):

- Wordbank 12
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Worksheets (11):

12.01 Games of chance
12.02 Coins probability • Dice probability • Probability review • Relative frequencies • Final 6 investigation
12.03 Venn diagrams
12.04 Two-way tables • Two-way probability tables
12.05 Tree diagrams
Mind map Probability

Puzzles (4):

12.01 Matching probabilities
12.03 Venn diagrams matching activity • Venn diagrams
Language of maths Probability crossword

Presentation (1):

12.05 Combined events

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- calculate the theoretical and observed probabilities of events, including complementary events
- (PATH) describe compound events using terminology such as 'and', exclusive 'or', inclusive 'or' and 'at least'
- (PATH) use Venn diagrams and two-way tables to represent sample spaces and compound events to solve probability problems
- use tree diagrams and tables (arrays) to represent the sample space of two-stage chance experiments, with and without replacement, to solve probability problems

Quiz
SkillCheck 12

SkillCheck ANSWERS ON P. 602

1 A bag contains 3 green lollies, 4 red lollies, 5 white and 8 yellow lollies.

a What fraction of the lollies are red?

b What percentage of the lollies are yellow?

c What percentage of the lollies are not yellow?

d What fraction of the lollies are green or white?

2 If a die is rolled, which one of the following events is most likely?
Select the correct answer **A**, **B**, **C** or **D**.

A an even number
B a number greater than 2
C a number greater than 3
D a number greater than 4

3 A card is drawn at random from a deck of cards. What is the probability that it is:

a a 7?
b a heart card?
c a black card?

4 Match each description (**a** to **f**) below with the correct set of all whole numbers (**A** to **F**) that meet that description.

a between 3 and 10
b at least 1
c more than 4
d at most 3
e 5 or less
f not more than 4
A {1, 2, 3, 4, 5}
B {4, 5, 6, 7, 8, 9}
C {1, 2, 3}
D {1, 2, 3, 4}
E {1, 2, 3, ...}
F {5, 6, 7, ...}

5 A die is rolled. Find the probability that the number that comes up is:

a 1
b more than 2
c odd
d composite

6 **a** Write the sample space for this spinner.

b Find, as a percentage, the probability that the number spun is:

i 5
ii at least 5
iii 5 or less
iv more than 5



Probability

12.01

The language of probability

A **chance experiment** is an activity or process that involves chance, for example, rolling a die or tossing a coin.

An **outcome** is the result of a chance experiment. For example, when rolling a die, there are 6 possible outcomes – the numbers 1 to 6.

The **sample space** is the set of all possible outcomes of an experiment. When rolling a die, the sample space is {1, 2, 3, 4, 5 and 6}.

An **event** is one or more outcomes of an experiment. When rolling a die, one event may be rolling an odd number, which consists of the 3 outcomes 1, 3 and 5.

A **trial** is one go or run of an experiment, for example, one roll of the die.

In a **random** experiment, every possible outcome has an equal (fair) chance of occurring.

Worksheet
Games of chance

Puzzle sheet
Matching probabilities

Videos
Probability

Why do share prices change?

Videos
The card counter
The prisoner's dilemma
The Monty Hall problem

Probability of an event

$P(E)$ means 'the **probability** of an event, E (occurring)'. If all possible outcomes are **equally likely**, then:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

or $P(E) = \dfrac{\text{number of outcomes matching } E}{\text{number of outcomes in the sample space}}$

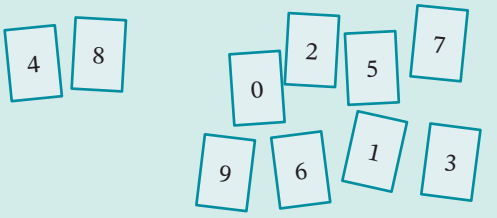
A **favourable outcome** is one of the outcomes in the event that you want, whose probability you are calculating.

Example 1

A card is chosen at random from a set of cards numbered 0 to 9.

Find the probability of selecting:

a the card numbered 8
b a number less than 4
c a number 5 or more



Solution

There are 10 outcomes in the sample space.

a $P(8) = \frac{1}{10}$ — 1 chance in 10

b Out of the 10 numbers, there are 4 numbers less than 4: {0, 1, 2, 3}.

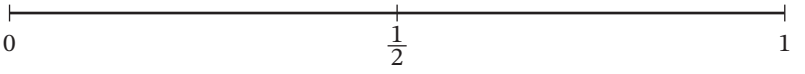
$P(\text{less than } 4) = \frac{4}{10} = \frac{2}{5}$ — 4 chances in 10

c Out of the 10 numbers, there are 5 numbers that are 5 or more: {5, 6, 7, 8, 9}.

$P(5 \text{ or more}) = \frac{5}{10} = \frac{1}{2}$ — 5 chances in 10

The range of probability

The probability of an event can be written as a fraction, decimal or percentage. Probability is measured on a scale from 0 to 1, or as a percentage from 0% to 100%. If an event has a probability of 0, it is **impossible** and it will not occur. If an event has a probability of 1, it is **certain** and it *must* occur.



impossible event
(For example, rolling a 7 on a die)

even chance
(For example, tossing tails on a coin)

certain event
(For example, the Sun will rise in the east)

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Complementary events

Video
Complementary events

Complementary events are 2 events that together make up all the possible outcomes, such as a head and a tail when tossing a coin. The **complement** of an event E is made up of those outcomes in the sample space that are 'not E', usually written as $\overline{E}$. Because an event and its complement covers all possible outcomes in the sample space, the sum of their probabilities must equal 1.

12.01

Complementary events

$P(\text{event}) + P(\text{complementary event}) = 1$

$P(E) + P(\overline{E}) = 1$

$P(\overline{E}) = 1 - P(E)$, where $\overline{E}$ is the complement of E.

Example 2

a List the sample space when this spinner is spun.

b Find $P(\text{red})$.

c What is the complementary event to the spinner landing on red?

d List the outcomes in the complementary event.

e Find $P(\text{not red})$.



Solution

a Sample space = {red, yellow, blue, green, white}

b $P(\text{red}) = \frac{1}{5}$

c The complementary event to landing on red is landing on a colour other than red.

d The outcomes in the complementary event are {yellow, blue, green, white}.

e $P(\text{not red}) = 1 - P(\text{red})$

$= 1 - \frac{1}{5}$

$= \frac{4}{5}$

EXERCISE 12.01 ANSWERS ON P. 602

Probability U F PS R C

EXAMPLE 1

1 A letter is drawn at random from the alphabet. PS R

a How many outcomes are there in the sample space?

b Are all the outcomes equally likely?

c What is the probability of choosing a vowel?

d What is the probability of choosing a letter that is in the word PROBABILITY?

Foundation Standard Complex

2 This spinner is spun. What is the decimal probability that the needle points to:

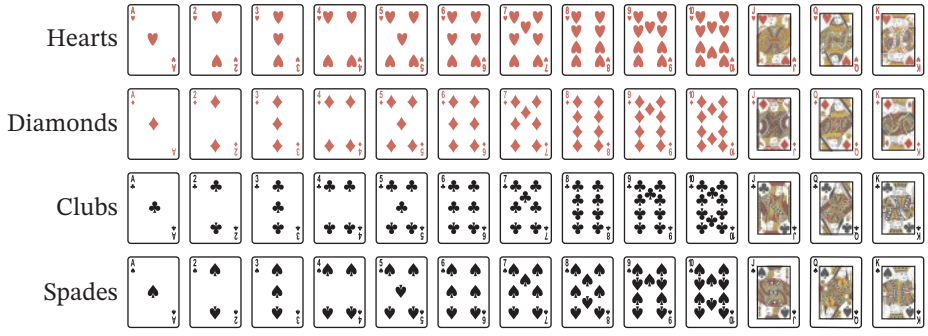


a yellow?

b black?

c red?

3 A normal pack of playing cards is divided into 4 suits as shown. PS R



A card is selected at random. Find the probability that it is:

a a black card

b an ace

c an even-numbered card

d a picture card

4 A die is rolled. What is the probability of rolling a number that is: PS R

a greater than 2?

b prime?

c a factor of 6?

d less than 4?

e greater than 6?

f less than 7?

5 A bag contains 15 red jelly beans and 20 blue jelly beans. One jelly bean is drawn at random from the bag. What is the probability it will be:

a blue?

b yellow?

c red or blue?

6 A student is selected at random from this group: Robert, Samantha, James, Joshua, Hayley, Jasmine, Caitlin, Dianne, Sarah. What is the probability of selecting a student: PS R

a who is male?

b who is female?

c whose name begins with J?

d whose name has one syllable?

e whose name contains an E?

7 A bag contains 14 black pens and 9 red pens. One pen is drawn at random from the bag. Calculate, as a percentage, the probability that it is:

a black

b red

c black or red

d blue

8 A die with faces marked 1, 1, 2, 3, 4 and 5 is rolled. Find the probability of rolling:

a 1

b 5

c an odd number

d a number greater than 1

Foundation Standard Complex

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9 A bag contains 35 blue badges, 45 green badges and 20 yellow badges. A badge is drawn at random. Calculate, as a percentage, the probability that it is:

a blue **b** green

c not green **d** neither blue nor yellow

10 For each event, write the complementary event. R C

a Selecting a consonant from the alphabet

b The weather being sunny tomorrow

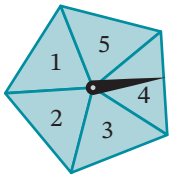
c A traffic light showing red

d Selecting a spade from a normal deck of cards

e Hitting the bullseye on a dartboard

f Selecting a blue ball from a bag containing blue, red and yellow balls

11 This spinner is spun. R C



a List the sample space.

b Find $P(4)$.

c What is the complementary event to spinning a 4?

d List the outcomes in the complementary event.

e If $P(\overline{4})$ means $P(\text{not } 4)$, find $P(\overline{4})$.

12 Dilone buys a ticket in a raffle in which 50 tickets are sold and there is only one prize.

a What is Dilone's chance of winning the prize? Write your answer as a:

i fraction **ii** decimal **iii** percentage

b Calculate the probability of Dilone not winning the prize. Write your answer as a:

i fraction **ii** decimal **iii** percentage

13 Tracey buys tickets in a raffle in which 200 tickets are sold and there is only one prize. She has a 40% chance of winning the prize. PS R

a What is the probability that Tracey will not win the prize?

b How many tickets did Tracey buy?

14 Tamin bought a ticket in a raffle in which 500 tickets were sold. If there are 5 prizes, what is the probability that Tamin does not win a prize? Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{1}{100}$ **B** 49% **C** $\frac{499}{500}$ **D** 0.99

15 The probability of rain on Tuesday is 0.35. What is the probability that it will not rain?

16 What is the probability that a student chosen at random from your school was not born in a month beginning with the letter M?

17 Stephanie has a 76% chance of hitting the bullseye when playing darts. What is the probability that she misses the bullseye? Select **A**, **B**, **C** or **D**.

A 24% **B** 26% **C** 34% **D** 76%

18 What is the probability that a mobile phone number selected at random does not begin with a 0?

Did you know?

The history of playing cards





Bridgeman Images/DeAgostini Picture Library/A. Dagli Orti

The earliest European cards were Tarot packs, which contained either 78 or 97 cards, and were used to play a game called Tarocco. The smaller 52-card pack may have become popular for practical reasons because cards were originally drawn and painted by hand. The names and symbols on cards have changed many times but in the 14th century a pack was introduced with the symbols for hearts, spades, diamonds and clubs, the symbols which were eventually adopted by most Western countries. The names for the symbols still differ from country to country.

The pictures used for the Kings, Queens and Jacks have undergone some changes. The four Kings originally represented Charlemagne (hearts), the emperor of most of Europe until 814 CE, David (spades), the Hebrew king mentioned in the Bible, Julius Caesar (diamonds) of ancient Rome and Alexander the Great (clubs) of ancient Greece.

Investigate who the original Queens and Jacks were.

12.02 Observed probability

Videos
Observed and expected frequencies
Experimental probability
Freak waves

Worksheets
Coins probability
Dice probability
Probability review
Relative frequencies
Final 6 investigation

Probability calculated using the formula:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

is more specifically called the **theoretical probability** or **expected probability**.

We can also determine probability based on the results of a trial that has been repeated many times, such as testing the effectiveness of 100 light globes, or rely on past statistics, such as weather patterns or the ages of drivers having car accidents. This type of probability is called **observed probability** or **experimental probability**, which is based on **relative frequency**, the number of times an event occurred as a fraction of the total frequency of outcomes.

Observed probability

$$P(E) = \frac{\text{number of times the event happened}}{\text{total number of trials}}$$

$$\text{or } P(E) = \frac{\text{frequency of } E}{\text{total frequency}}$$

Expected frequency is the expected number of times an event will occur over repeated trials.

Expected frequency

Expected frequency = theoretical probability × number of trials.

Example 3

Connor tossed a coin 100 times and recorded the results in a table.

Outcome	Frequency
Head	46
Tail	54
	100

a Calculate, as a decimal,

i the observed probability of tossing a head

ii the theoretical probability of tossing a head.

b Are the observed and theoretical probabilities similar?

c For 100 tosses, what is the expected frequency of heads based on the theoretical probability? How does this compare with the observed frequency?

Solution

a **i** Observed probability: $P(H) = \frac{46}{100} = 0.46$

ii Theoretical probability: $P(H) = \frac{1}{2} = 0.5$

b By comparing the decimals for the 2 answers, we see that the observed and theoretical probabilities are similar.

c Expected frequency of heads $= 0.5 \times 100$ probability × number of trials

$= 50$

From the table, the observed frequency of heads = 46, which is close to 50.

Note: When calculating observed probability, the results will not always be consistent with what is expected, unless a very large number of trials are performed.

Example 4

Briana rolled a die 200 times and recorded her results.

Outcome	Frequency
1	38
2	40
3	29
4	24
5	35
6	34

a Find the observed probability of rolling:

i an even number

ii an even number or a number greater than 2

iii an even number and a number greater than 2

iv an even number not greater than 2.

b Calculate the probability of rolling a 5 or 6

i as an observed probability **ii** as an expected probability.

c What is the expected frequency of rolling a 5 or 6 in 200 rolls of a die? How does this compare with Briana's observed frequency?

Solution

a **i** Rolls of even numbers $= 40 + 24 + 34$ Frequencies of 2, 4, 6

$= 98$

Observed $P(\text{even}) = \frac{98}{200} = \frac{49}{100}$

ii Rolls of even numbers or numbers greater than 2 $= 40 + 24 + 34 + 29 + 35$ Frequencies of 2, 4, 6, 3, 5

$= 162$

Observed $P(\text{even or} > 2) = \frac{162}{200} = \frac{81}{100}$

Interactive
Plinko probability

12.02

iii Rolls of even numbers greater than 2 — Frequencies of 4 and 6

$= 24 + 34$

$= 58$

Observed $P(\text{even and} > 2) = \frac{58}{200} = \frac{29}{100}$

iv Rolls of even numbers not greater than 2 $= 40$ — Frequency of 2

Observed $P(\text{even and not} > 2) = \frac{40}{200} = \frac{1}{5}$

b **i** Rolls of 5 or 6 $= 35 + 34$ — Frequencies of 5 and 6

$= 69$

Observed $P(5 \text{ or } 6) = \frac{69}{200}$

ii Expected $P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$ — 2 chances out of 6

c Expected frequency of 5 or 6 $= \frac{1}{3} \times 200$ — probability × number of trials

$= 66.666...$

≈ 67

From the table, the observed frequency $= 35 + 34 = 69$, which is close to 67.

EXERCISE 12.02 ANSWERS ON P. 603

Observed probability U F R C

EXAMPLE 3

1 Fatima spun this spinner 200 times and recorded the results in the table. R C



Event	Frequency
Red	90
Green	36
Blue	74
	200

a Calculate, as a decimal, the observed probability that the needle points to:

i red **ii** green

b Calculate, as a decimal, the theoretical probability that the needle points to:

i red **ii** green

c Are the observed and theoretical probabilities similar?

d For 200 spins, what is the expected frequency of red or blue based on the theoretical probability? How does this compare with the observed frequency?

2 Use cardboard and a toothpick to make a square spinner that has 4 colours as shown in the diagram below. Spin the spinner 50 times and record your results in a table similar to the one shown. R C



Event	Tally	Frequency
Red		
Blue		
Green		
Yellow		

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a Calculate, as a percentage, the relative frequency of spinning:

i red **ii** blue

iii a colour that is not yellow **iv** blue or green

b Calculate, as a percentage, the expected probability of spinning blue. How does this compare with the relative frequency?

c For 50 spins, what is the expected frequency that blue should come up? How does this compare with the observed frequency?

3 A die is rolled. R C

Event	Tally	Frequency
1		
2		
3		
4		
5		
6		
		100

a Predict the number of times a 2 would appear if the die was rolled 100 times.

b Roll a die 100 times and record your results in a table similar to the one shown.

c What is the relative frequency of rolling a 2?

d How does your predicted frequency compare with the observed frequency?

e What is the relative frequency of rolling an even number?

4 Place 4 black counters, 5 red counters and 1 yellow counter in a bag. Select a counter at random, record the colour and return the counter to the bag. Run this experiment 50 times. R C

Event	Tally	Frequency
Black		
Red		
Yellow		

a What is the experimental probability of selecting:

i a black counter?

ii a red counter?

iii a yellow counter?

b What is the theoretical probability of selecting:

i a black counter?

ii a red counter?

iii a yellow counter?

c Compare your experimental probabilities to the theoretical probabilities.

d Repeat the experiment another 50 times and find the relative frequency of each event. Are the results different to the results found in part **a**?

EXAMPLE 4

5 A die was repeatedly rolled and the results are shown in the table.

Outcome	Frequency
1	90
2	84
3	92
4	76
5	83
6	75

a What was the total number of trials?

b Find the experimental probability (as a decimal) of rolling:

i an even number

ii a number greater than 3

iii a 3 or a 5

iv a number greater than 2 and even

vii a number greater than 3 or odd.

Foundation Standard Complex

c Calculate the probability of rolling a number not less than 3:

- **i** as an experimental probability
- **ii** as an expected probability.

d How do the 2 probabilities in part **c** compare?

6 Students at a high school were asked how they travelled to school. The results are presented in the table. Find the probability (correct to 2 decimal places) that a student chosen at random will:

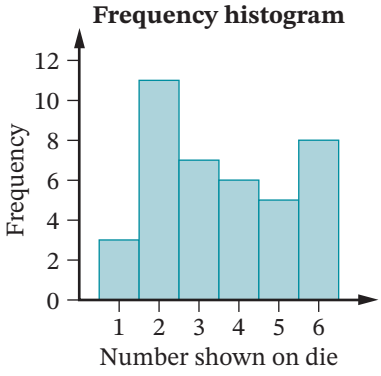
Outcome	Frequency
Walk	186
Bus	295
Car	103
Bicycle	74

a walk **b** be driven

c catch a bus **d** not ride a bicycle

e be driven or catch a bus.

7 A die was rolled many times and the results are shown in the frequency histogram.



a What is the total number of trials?

b Find the observed probability of rolling:

- **i** 2
- **ii** 5
- **iii** an even number or a number greater than 3.
- **iv** a number greater than 4 and odd.

8 The table shows the number of babies born per day at a hospital during the month of January. On any particular day, find the probability of the birth of exactly:

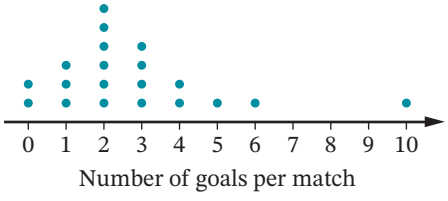
Number of babies per day	Frequency
0	1
1	4
2	7
3	10
4	5
5	4

a 1 baby **b** 3 babies

c 1 or 2 babies **d** **at least** 2 babies

e at most 1 baby **f** 4 or 5 babies.

9 This dot plot shows the number of goals per match scored by a soccer team during the season.



For a match selected from the season at random, find the probability of the team scoring:

a 2 goals **b** 4 goals per game **c** fewer than 2 goals

d 2 or 3 goals **e** at least 3 goals **f** not more than 4 goals

10 A card is drawn randomly from a normal pack of cards to see if it is an Ace. This trial is run 120 times, with each card being replaced in the deck before the next card is drawn. The results are shown in the table. R C

Outcome	Frequency
Ace	11
Not an Ace	109

a Calculate, correct to 3 decimal places:
- **i** the observed probability of selecting an Ace
- **ii** the theoretical probability of selecting an Ace.

b Are the observed and theoretical probabilities similar?

c If a card is drawn 300 times, what is the expected frequency of an Ace being drawn? Select the correct answer **A**, **B**, **C** or **D**.

A 27 **B** 28 **C** 23 **D** 6

12.02

Technology

Throwing a coin

Worksheet
Coins probability

In this activity, a spreadsheet is used to simulate the throwing of a coin. The computer can quickly generate either a 1 (representing heads) or a 2 (tails).


Shutterstock.com/Yeti studio

1. In cell A1, enter the label 'Tossing a coin'.
2. In cell A2, enter the formula **=INT(RAND()*2+1)**, press Enter and either 1 or 2 will appear in the cell.
3. Select cell A2 and **Fill Down** to cell A11 to simulate the tossing of a coin 10 times.
4. Copy the table and in the first blank row record your results for Trials 1–10 (the numbers of heads and tails in your first 10 'tosses').
5. Simulate another 10 tosses of the coin by making the spreadsheet generate another set of random numbers. Do this by placing the cursor between the tops of columns A and B (so that it turns into a 'double arrow') and clicking.
6. Enter the results for Trials 11–20 in your table.
7. Repeat 8 more times so that you have 100 tosses of the coin recorded in the table.
8. **a** For 100 tosses of a coin, what is the expected frequency of heads based on the theoretical probability?
   b Compare this with the actual frequency.
9. Calculate, as a decimal, the theoretical and observed probabilities of tossing heads.
10. Compare your results with those of other students in your class. Briefly explain any differences and why they may have occurred.
11. Combine the results of students in your class to calculate the observed probability of throwing heads, as a decimal. How does this compare with the theoretical probability?

Trial	Number of heads	Number of tails
1–10		
11–20		
21–30		
31–40		
41–50		
51–60		
61–70		
71–80		
81–90		
91–100		
Total		

Foundation Standard Complex

Worksheet
Dice probability

Technology

Rolling dice

In this activity, we will use a scientific calculator to simulate the rolling of a die.

1 Copy the table shown.

Outcome	Tally	Frequency	Observed probability
1			
2			
3			
4			
5			
6			

2

Casio scientific calculator	Sharp scientific calculator
Enter the formula RanInt#(1,6) as shown: ALPHA . 1 SHIFT) 6) =	2nd F 7 (RANDOM), select 1 R-DICE, =

This generates a random number from 1 to 6. Press = more times to generate more random numbers. Alternatively, on a spreadsheet, type **=INT(RAND()*6+1)** into a cell and press the Enter key, then the F9 key repeatedly for more random numbers.

3 Run the simulation 50 times and record the results in the table.

4 Calculate the theoretical probability for each number being tossed on the die.

5 Compare the theoretical probability and observed probability for each number. Are there similarities or differences? How could the observed probabilities be more accurate? Explain.

6 Copy this table for rolling 2 dice and finding their sum.

Sum	Tally	Frequency	Observed probability
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

7 On a scientific calculator, rolling 2 dice is achieved by adding the result of rolling one die twice.

Casio scientific calculator	Sharp scientific calculator
Enter the formula RanInt#(1,6) + RanInt#(1,6) as shown below: ALPHA · 1 SHIFT) 6) + ALPHA · 1 SHIFT) 6) =	Enter the formula R.DICE + R.DICE as shown below: 2nd F 7 (RANDOM), select 1 R-DICE, + 2nd F 7 (RANDOM), select 1 R-DICE, =

This generates the sum of 2 random numbers from 1 to 6.

Press = more times to generate more sums.

8 Run the simulation 50 times and record the results in the table.

9 Which sum is the most likely to occur?

10 Which sum is the least likely to occur? Why?

11 Which sum is theoretically the:

a most likely? **b** least likely?

Why?

12 Compare your answers to questions **9**, **10** and **11** and then compare them with the answers of other students in your class. Write a few sentences explaining your observations.

13 If you ran 50 more simulations, would you expect your results to be similar or different to your first 50 results? Explain.



Shutterstock.com/Malochka Mikalai

12.02

12.03 Venn diagrams

PATH

Videos
Venn diagrams 2

Venn diagrams: Global habitats

Worksheet
Venn diagrams

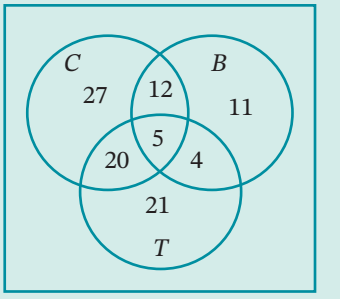
Puzzles
Venn diagrams matching activity

Venn diagrams

A **Venn diagram** is a diagram of circles (usually overlapping) for grouping items into categories. A rectangle represents the whole group, while the circles represent categories. Items common to 2 or more categories are placed in the intersection (overlapping region) of the circles. The Venn diagram was invented in 1880 by English mathematician and priest, John Venn (1834–1923).

Example 5

This Venn diagram shows the results of a survey on how people travel to work (car (C), bus (B) or train (T)).



a How many people participated in the survey?

b How many people used all 3 methods: car, bus *and* train? Give an explanation on how this might happen.

c Calculate the probability of selecting a person from this survey who travels by:

i car only or bus only **ii** car and train

iii car and train only **iv** bus but not train.

d A person is chosen at random from the people who travel by car or train only (but not bus). What is the probability that the person travels by car only?

e If 110 people actually participated in the survey, what extra information would be needed on the diagram?

Solution

a Number of people $= 27 + 12 + 5 + 20 + 21 + 4 + 11$

$= 100$

b 5 people travelled by car, bus and train.

The region where the 3 circles intersect

They may have driven to the station, caught a train and then a bus to the workplace, or used different methods on different days.

c **i** People travelling by car only or bus only $= 27 + 11$

$= 38$

The regions in C and B that don't overlap

$P(\text{car only or bus only}) = \frac{38}{100}$

'Car only or bus only' is an example of a **compound event**.

$= \frac{19}{50}$

ii People travelling by car and train $= 20 + 5$

$= 25$

The region where C and T intersect

$P(\text{car and train}) = \frac{25}{100}$

'Car and train' is another example of a **compound event**.

$= \frac{1}{4}$

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iii People travelling by car and train only = 20 — Where C and T intersect, but excluding B

$P(\text{car and train only}) = \frac{20}{100}$

$= \frac{1}{5}$

iv People travelling by bus but not train — The circle B excluding T

$= 11 + 12$

$= 23$

$P(\text{bus but not train}) = \frac{23}{100}$

d People travelling by car or train only $= 27 + 20 + 21$

$= 68$

People travelling by car only = 27.

$P(\text{car only}) = \frac{27}{68}$

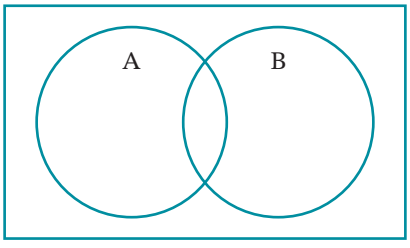
e 110 – 100 = 10. This means that there were 10 people who did not use a car, bus or train. A '10' would need to be placed in the rectangle, but outside of the three circles.

When we combine 2 or more simple events, we get a **compound event**. In the above example, 'car and train' and 'bus but not train' are compound events.

'And' vs 'or'

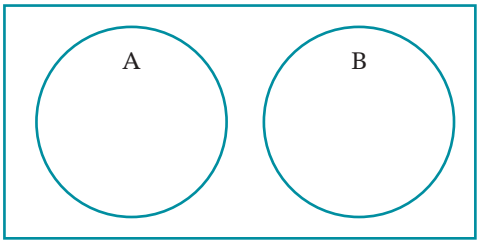
For 2 categories or events A and B, the compound event **'A and B'** means to have both of them occurring together. For example, 'to drive a car' **and** 'to ride a bus' means to do both things.

If A and B are **overlapping**, the compound event **'A or B'** means to have A or B or both. For example, 'to drive a car' **or** 'to ride a bus' means to drive a car only, or to ride a bus only, or to do both. In this case, 'A or B' actually **includes** 'A and B' so this is an example of an **inclusive** 'or'.



Overlapping events: 'A or B' means A or B or both

If A and B are **mutually exclusive**, this means that they are **not overlapping** and on a Venn diagram they appear as 2 separate circles. For mutually exclusive categories or events, the phrase **'A or B'** means to have A only or B only (but not both). For example, 'male' **or** 'female' means to be male, or female, but not both. In this case, 'A or B' **excludes** 'A and B' so this is an example of an **exclusive** 'or'.



Mutually exclusive events: 'A or B' means A or B but not both

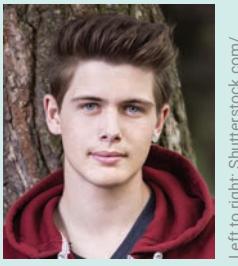
PATH

Video
Venn diagrams 1

Example 6

A survey of 110 students at Redford College showed that 61 students had dark hair, 42 students had brown eyes and 20 students had dark hair and brown eyes.





Left to right: Shutterstock.com/michaeljung, Alamy Stock Photo/blickwinkel, Shutterstock.com/Dorota Zietek

a Represent this information on a Venn diagram.

b How many students had brown eyes or dark hair but not both?

c What is the probability of randomly selecting from this group a student:

i with dark hair?

ii with dark hair and brown eyes?

iii with dark hair or brown eyes?

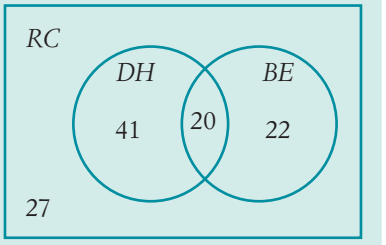
iv with neither dark hair nor brown eyes?

Solution

a RC = students at Redford College

DH = students with dark hair

BE = students with brown eyes



When completing a Venn diagram, always begin with the intersection.

There are 20 students with dark hair and brown eyes.

$\therefore$ Students with dark hair only $= 61 - 20$

$= 41$

$\therefore$ Students with brown eyes but not dark hair $= 42 - 20$

$= 22$

$\therefore$ (Remaining) Students with neither dark hair nor brown eyes $= 110 - 41 - 20 - 22$

$= 27$

b Number of students with brown eyes or dark hair only $= 22 + 41$

$= 63$

c **i** Students with dark hair $= 41 + 20$

$= 61$

$P(\text{dark hair}) = \frac{61}{110}$

ii $P(\text{dark hair and brown eyes}) = \frac{20}{110}$

$= \frac{2}{11}$

'Dark hair and brown eyes' is a compound event.

iii Number of students with dark hair or brown eyes $= 41 + 20 + 22$

$= 83$

$P(\text{dark hair or brown eyes}) = \frac{83}{110}$

iv $P(\text{neither dark hair nor brown eyes}) = \frac{27}{110}$

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EXERCISE 12.03 ANSWERS ON P. 603

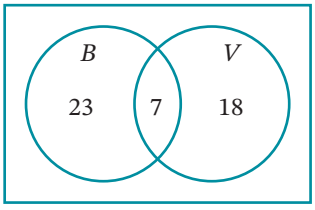
PATH

Venn diagrams U F R C

EXAMPLE 5

12.03

1 The Venn diagram shows the number of students who play either basketball (B) or volleyball (V). R



a Find the total number of students who play basketball or volleyball.

b Find the probability of randomly selecting a student who plays:

i basketball **ii** volleyball.

c What is the probability of selecting a student who plays:

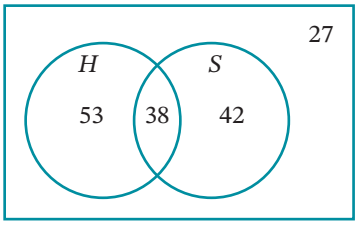
i basketball or volleyball? **ii** basketball and volleyball?

iii basketball but not volleyball? **iv** neither basketball nor volleyball?

d Of the students who play volleyball, find the probability of selecting a student who also plays basketball. Select the correct answer **A**, **B**, **C** or **D**.

A 0.28 **B** 0.375 **C** 0.72 **D** 0.92

2 The Venn diagram shows the number of Year 9 students who achieved Credit grades in their Science (S) and History (H) exams. R



a How many students are in Year 9?

b Find the probability of selecting a student who achieved a Credit in:

i History **ii** Science

iii History and Science **iv** History or Science

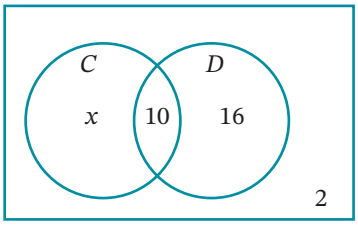
v History or Science but not both **vi** History only

vii Science only

c What is the percentage probability of selecting a student who did not achieve a Credit in both History and Science?

d Out of the students who achieved a Credit in History, what is the probability that the student also achieved a Credit in Science?

3 40 people were asked whether they preferred watching movies at the cinema (C) or on a device (D) at home. The results are shown in the Venn diagram. R C



a Describe the category that contains the pronumeral x in the diagram.

b Find the value of x.

c Calculate the decimal probability of randomly selecting a person who likes:

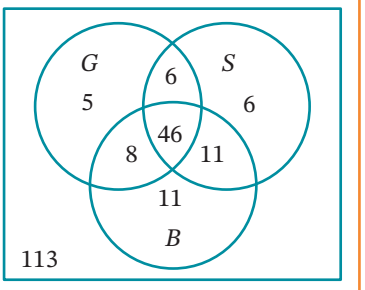
i the cinema and a device **ii** a device only

iii the cinema or a device **iv** neither the cinema nor a device.

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PATH

4 This Venn diagram describes the number of countries that won gold, silver, bronze or no medals at the (postponed) 2021 Olympic Games in Tokyo, Japan. R C



a How many countries competed at these games? Select **A**, **B**, **C** or **D**.

A 93 **B** 177

C 206 **D** 218

b How many countries did not win a medal? How is this shown on the Venn diagram?

c What is the probability of randomly selecting a country that:

i won only gold medals?

ii won gold or silver medals, but not both?

iii won gold, silver and bronze medals?

iv won gold and bronze only?

v won gold only or silver only or bronze only?

vi did not win a gold medal?

vii did not win a gold or silver medal?

d Out of the countries that won medals, what is the probability of selecting a country that:

i won gold medals? **ii** won silver but not bronze?



Alamy Stock Photo/PA Images

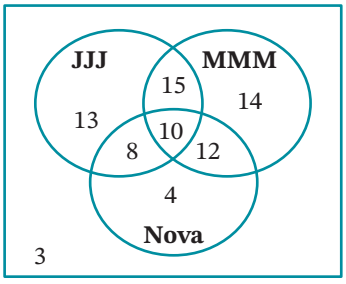
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PATH

12.03

5 A group of teachers were asked what radio stations they listened to and the results are shown in the Venn diagram. R



a How many teachers were surveyed?

b What is the probability of selecting a teacher who did not listen to the 3 stations?

c What is the probability of selecting a teacher who listens to:

i JJJ? **ii** MMM? **iii** Nova?

d What is the probability of selecting a teacher who listens to at least 2 of the stations? Select **A**, **B**, **C** or **D**.

A $\frac{50}{79}$ **B** $\frac{45}{79}$ **C** $\frac{35}{79}$ **D** $\frac{30}{79}$

e Find the probability of selecting a teacher who listens to:

i JJJ or MMM but not both

ii all 3 stations

iii MMM only

EXAMPLE 6

6 In a drama group of 30 people, 18 people dance (D), 20 people sing (S) and 8 people dance and sing. R C

a Show this information on a Venn diagram.

b Find the probability of selecting a person who:

i dances and sings

ii sings but does not dance

iii dances but does not sing

iv dances or sings

v dances or sings but not both.

7 A class of 30 students were given the option of playing an indoor sport (S) or watching a movie (M) when regular sport was cancelled due to bad weather. 15 students wanted to play an indoor sport and 24 students wanted to watch a movie. R C

a How many students indicated that they would do either?

b Copy and complete a Venn diagram using the information given.

c For this Venn diagram, there are no values outside the circles. Explain why.

d What is the probability of selecting a student who prefers to:

i play an indoor sport or watch a movie, but not both?

ii watch a movie but not play an indoor sport?

iii play an indoor sport but not watch a movie?

PATH

8 80 families living in Mt Robinson were surveyed on whether they had a dog (D) or a cat (C) as a pet. 12 families had no pets, 35 had a dog and 47 had a cat. R C



Shutterstock.com/Chendongshan

a Draw Venn diagram showing the information given.

b How many families had:

i a dog and a cat?

ii a dog but not a cat?

iii at least a dog or a cat?

iv a dog or a cat, but not both?

c Find the decimal probability of selecting a family with:

i a dog only

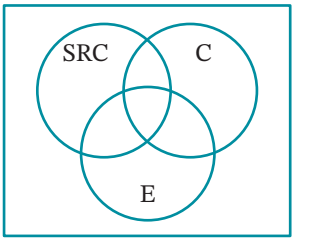
ii either a dog or a cat, but not both

iii at most one pet

iv at least one pet.

d Given that a family has a pet, what is the probability that it is a cat?

9 At Nelson River High School, the Student Representative Council (SRC) has 25 members, the Chess club has 20 members and the Environment club has 33 members. 5 students belong to all three groups, while 13 members belong to the SRC and no other club. 4 members of the Environment club are members of the SRC, but not the Chess club, while 10 members belong to both the Environment and Chess clubs. There are 8 members of the Chess club who are members of the SRC. R C



a Copy and complete the Venn diagram.

b How many students belong to the Environment club only?

c How many students belong to the Chess club and no others?

d Find the probability of selecting a student who belongs to exactly 2 clubs.

e What is the probability of a student belonging to at least 2 clubs?

f Find the probability of selecting a student who belongs to:

i the Environment club and the SRC, but not the Chess club

ii only one club

iii the SRC or the Chess club, but not the Environment club

iv at most 2 clubs

g Is it necessary to include the rectangle in this Venn diagram? Give reasons.

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Two-way tables

12.04

A **two-way table** is another way of grouping items into overlapping categories, especially when there are many overlaps that cannot easily be represented by Venn diagrams.

PATH

12.04

Video
Two-way tables 2

Worksheets
Two-way tables

Two-way probability tables

Example 7

This table categorises the heights and ages of 200 Year 9 students.

	Short	Tall
Aged 14	45	25
Aged 15	90	40

a How many students are:

i aged 15 and tall?

ii aged 14 or short?

b What fraction of students are neither tall nor aged 15?

c What is the probability of selecting a student at random who is:

i aged 14 and tall?

ii aged 15 but not tall?

iii tall or aged 14, but not both?

d If a short student is selected at random, what is the probability that he or she is aged 14?

Solution

a i Students aged 15 and tall = 40

ii Students aged 14 or short = 45 + 25 + 90

= 160

b Students who are neither tall nor 15 = 45

$\therefore$ Fraction of students neither tall nor 15 $= \frac{45}{200} = \frac{9}{40}$

The short 14-year-olds

c i $P(\text{tall and 14}) = \frac{25}{200}$

$= \frac{1}{8}$

'Tall and 14' is a compound event.

ii $P(\text{15 but not tall}) = \frac{90}{200}$

$= \frac{9}{20}$

iii Students tall or aged 14, but not both = 40 + 45

= 85

$P(\text{tall or 14, but not both}) = \frac{85}{200} = \frac{17}{40}$

d Short students = 90 + 45 = 135

Students who are short and 14 = 45

$P(\text{short student who is 14}) = \frac{45}{135} = \frac{1}{3}$

PATH

EXERCISE 12.04 ANSWERS ON P. 604

Two-way tables U F R C

EXAMPLE 7

1 Households were surveyed at random to see whether or not they had dishwashers and clothes dryers. The results are in the table. R C

	Dishwasher	No dishwasher
Clothes dryer	75	108
No clothes dryer	49	68

a How many households were surveyed?

b How many households had a clothes dryer and a dishwasher?

c How many households had neither?

d What is the probability of selecting a household at random that has:

i a dishwasher?

ii a dishwasher and a dryer?

iii a dishwasher or a dryer, but not both?

iv at least one of these?

2 People at a train station were asked whether they had eaten breakfast or lunch that day. R C

	Ate lunch	Did not eat lunch
Ate breakfast	35	26
Did not eat breakfast	54	15

a How many people were surveyed?

b What percentage of people ate lunch?

c Find the probability of picking a person at random who ate:

i breakfast

ii lunch

iii breakfast but not lunch

iv breakfast or lunch

v breakfast and lunch

vi breakfast or lunch, but not both

d What is the probability of picking a person who has eaten lunch given that the person ate breakfast? Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{26}{61}$ **B** $\frac{35}{61}$ **C** $\frac{35}{89}$ **D** $\frac{89}{130}$

e What is the probability of picking a person who has not eaten lunch given that the person did not eat breakfast?

3 People at a shopping mall were asked if they watched football on TV. R C

	Football	No football
Male	85	27
Female	36	92

a How many people were surveyed?

b How many people:

i watched football?

ii did not watch football?

c What percentage of people surveyed were females?

d What is the probability of randomly selecting a person who did not watch football and was male?

e Given that a person watched football, what is the probability that the person was female?

4 The results comparing gender and handedness of students at a primary school are shown in the table. R C

	Male	Female
Left-handed	14	18
Right-handed	86	65

a How many students are at the school?

b What is the probability of selecting a student at random who is:

i male and left-handed?

ii male or left-handed?

c Find the probability, expressed as a percentage, of randomly selecting a right-handed female.

d Given that a student is a male, what is the probability that he is right-handed?

e What is the probability of selecting a female or a right-handed student?

5 After a Federal election, the members (politicians) of the House of Representatives in the Australian Parliament were as shown. R C

	Government	Opposition
Male	62	43
Female	15	31

a How many members of parliament (MPs) were there in the House of Representatives?

b Find the percentage probability of randomly selecting an MP who is:

i male

ii female and in Opposition

iii in the Government or female

iv male and in Opposition.

c Find the probability of selecting an MP who is male and in Government.

d If a female MP is selected at random, what is the probability that she is in Government?



Getty Images/Tracey Nearmy

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PATH

6 Students at Nelson Anglican College were surveyed on whether they played video games. The results are shown in the table. R C

	Plays video games	Does not play video games
Female	178	267
Male	310	210

a How many students attend the college?

b Find the percentage probability that a student selected at random from the college:

i plays video games and is female

ii doesn't play video games

iii is male and doesn't play video games

iv is female and doesn't play video games.

c If a female student is selected at random, what is the probability that she plays video games?

d If a male student is selected at random, what is the probability that he plays video games?

7 A group of retired people was surveyed on whether they exercised regularly. R C

	Exercise	Don't exercise
Men	65	106
Women	74	75

a How many people were surveyed?

b What percentage of people exercised?

c What is the probability of selecting a person at random who doesn't exercise regularly?

d Of the women who were surveyed, what percentage exercised regularly?

e If a man was selected randomly from this survey, what is the probability that he exercises regularly? Select **A**, **B**, **C** or **D**.

A 38.0% **B** 62.0% **C** 20.3% **D** 53.4%

Investigation

Using two-way tables

Work in groups of 3 or 4.

Investigate 2 characteristics, for example, gender and exercise, gender and eating habits, eating habits and exercise, playing sport and going to the movies, eye colour and hair colour.

Conduct a survey to obtain the information required, then summarise your results in a two-way table similar to this one:

	Eat regular meals	Don't eat regular meals
Play sport		
Don't play sport		

Note any patterns in your data. This may include answering questions such as 'What percentage of ...' or 'What is the probability of picking a person who ...'.

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Mental skills 12: Maths without calculators ANSWERS ON P. 604

Quiz
Mental skills 12

12.04

The unitary method with percentages

The unitary method is used when you are only given a *percentage* of an amount and you need to find the amount. It is called the unitary method because we find 1% of the amount first, then multiply that by 100 to find the whole (100%).

1 Study each example.

a If 8% of a number is 24, what is the number?

8% of the number = 24

$\therefore$ 1% of the number = 24 ÷ 8 = 3

$\therefore$ 100% of the number = 3 × 100 = 300

The number is 300. *Check*: 8% × 300 = 24

b If 15% of an amount is \$90, what is the whole amount?

15% of the amount = \$90

$\therefore$ 1% of the amount = \$90 ÷ 15 = \$6

$\therefore$ 100% of the amount = \$6 × 100 = \$600

The amount is \$600. *Check*: 15% × \$600 = \$90

2 Find the whole amount if:

a 5% of the amount is \$35

b 11% of the amount is \$88

c 20% of the amount is 80

d 6% of the amount is 42

e 90% of the amount is \$270

f 15% of the amount is \$60

g 40% of the amount is 100

h 120% of the amount of \$360

i 25% of the amount is \$75

j 8% of the amount is 40

Did you know?

Lotto

Lotto is a game of chance in which 6 balls are selected from a barrel of balls numbered 1 to 45. Players who correctly predict the 6 numbered balls share a first prize of at least 4 million dollars. The original Lotto game introduced in 1979 used 40 balls, and by increasing to 44 balls in 1989, the chance of winning first prize dropped from 1 in 3 838 380 to 1 in 7 059 052. The addition of 4 extra balls made Lotto almost twice as difficult to win. Another ball was added in 2004, making the chance of winning first prize even slimmer.



iStock.com/askhamdesign

Research the probability of winning Lotto today now that it uses 45 balls.

Investigation

Are the Lotto numbers equally likely?

In each Lotto game, 8 numbers are selected at random from the barrel: 6 main numbers, plus 2 supplementary numbers. Between January 2015 and October 2019, there were 2000 Lotto numbers drawn on the Saturday game. The frequency of each number (from 1 to 45) were as follows:

1	53	10	50	19	36	28	36	37	38
2	37	11	53	20	45	29	49	38	47
3	39	12	41	21	43	30	47	39	47
4	41	13	44	22	53	31	42	40	35
5	53	14	41	23	43	32	48	41	39
6	44	15	47	24	47	33	45	42	45
7	46	16	51	25	39	34	45	43	35
8	41	17	41	26	52	35	39	44	45
9	40	18	57	27	52	36	46	45	43

a If 2000 numbers were drawn over the 5 years, and each number from 1 to 45 was equally likely, how often would you expect each number to come up?

b How many of the frequencies from the above table are equal to this expected number?

c Which Lotto number came up the:

i most often? **ii** least often?

d Do you think the Lotto numbers are equally likely, or is the Lotto draw biased? Give a reason for your answer.

Source: www.thelott.com

12.05 Tree diagrams

Presentation Combined events

A **two-stage experiment** or **two-step experiment** is a chance experiment that has 2 parts or stages, for example:

- rolling 2 dice
- tossing a coin and die together
- drawing 2 prizes in a raffle
- observing the weather each day over a weekend.

The outcome of the second stage may or may not be affected by the outcome of the first stage.
The sample space for a two-stage experiment can be displayed using lists, tables (also called arrays) or tree diagrams.

Video Tables and tree diagrams

Example 8

Two coins are tossed together.

a Find all the outcomes in the sample space:

i by simply listing them **ii** using a table.

b What is the probability of tossing:

i a head and a tail? **ii** 2 tails? **iii** at least one head?

Solution

a Let H = heads and T = tails.

i The possible outcomes are {HH, HT, TH, TT}. — Finding the 4 outcomes by listing

ii

		1st coin	
		H	T
2nd coin	H	HH	TH
	T	HT	TT

Using a table ensures that all outcomes are counted.

b **i** $P(\text{a head and a tail}) = P(\text{HT or TH}) = \frac{2}{4} = \frac{1}{2}$ — 2 chances out of 4

ii $P(\text{2 tails}) = P(\text{TT}) = \frac{1}{4}$ — 1 chance out of 4

iii $P(\text{at least one head}) = P(\text{1 or 2 heads})$ — 'At least one' means '1 or more'

$= P(\text{HT or TH or HH})$

$= \frac{3}{4}$ — 3 chances out of 4

Example 9

While staying at a hotel, John-Paul can choose one item from each course of a breakfast menu.

1st course	2nd course
Cereal (C)	Bacon and eggs (B)
Raisin toast (R)	Pancakes (P)
Watermelon (W)	Sausages and hash browns (S)
Yoghurt (Y)	

a Use a table to list all the different 2-course breakfasts available.

b If one of these combinations is chosen at random, find the probability that it includes:

i watermelon but not pancakes **ii** yoghurt or pancakes

iii yoghurt and pancakes.

'Watermelon, not pancakes' is a compound event.

Solution

a

		1st course			
		C	R	W	Y
2nd course	B	CB	RB	WB	YB
	P	CP	RP	WP	YP
	S	CS	RS	WS	YS

$4 \times 3 = 12$ different breakfasts possible.

b **i** $P(\text{watermelon, not pancakes}) = \frac{2}{12} = \frac{1}{6}$ — WB or WS

ii $P(\text{yoghurt or pancakes}) = \frac{6}{12} = \frac{1}{2}$ — column Y + row P

iii $P(\text{yoghurt and pancakes}) = \frac{1}{12}$ — YP

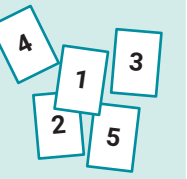
Worksheet
Tree diagrams

Videos
Tree diagrams 1
Tree diagrams 2
Tree diagrams 3

A **tree diagram** lists all the possible outcomes of each stage. Branches stretch out to show the possible pathways of outcomes at each stage. An outcomes column at the end of the diagram lists the sample space.

Example 10

From a set of cards numbered 1 to 5, two cards are drawn at random (without replacement) to make a 2-digit number.



a Show all outcomes in the sample space using a tree diagram.

b What is the probability of making a number:

i less than 32? **ii** divisible by 3?

Solution

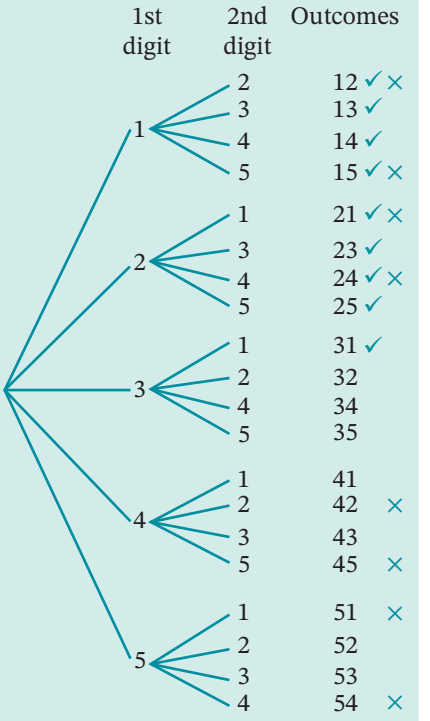
a This experiment has 2 stages: the 1st digit and the 2nd digit.

For the 1st digit, there are 5 possible digits: 1, 2, 3, 4 or 5.

For the 2nd digit, there are only 4 possible digits left because the 1st digit cannot be used again ('without replacement').

There are $5 \times 4 = 20$ possible 2-digit numbers in the sample space.

Using a tree diagram ensures that all outcomes are counted.



b **i** There are 9 numbers that are less than 32 (ticked in the tree diagram).

$\therefore P(\text{number} < 32) = \frac{9}{20}$.

ii There are 8 numbers that are divisible by 3 (crossed in the tree diagram).

$\therefore P(\text{number divisible by 3}) = \frac{8}{20}$

$= \frac{2}{5}$.

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EXERCISE 12.05 ANSWERS ON P. 604

Tree diagrams U F PS R C

EXAMPLE 8

12.05

1 2 people out of Ben, Anita, William, Lachlan and Noreen are to be selected for the positions of president and secretary in a tennis club. PS R C

a List the possible pairings for the positions of president and secretary.

b What is the probability of the committee consisting of:

i Ben and Anita?

ii 2 males?

iii a female president?

iv at least one female?



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2 The digits 5, 6, 8 and 9 are written on cards and put in a bag. 2 cards are drawn at random to make a 2-digit number. PS R C

a Make a list of all possible outcomes.

b How many 2-digit numbers are possible?

c Find as a decimal (**iv** to 2 decimal places) the probability that the number is:

i even

ii greater than 60

iii between 60 and 90

iv divisible by 3

EXAMPLE 9

3 A coin and a die are tossed together. PS R C

a Copy and complete the table to display the sample space.

		Die					
		1	2	3	4	5	6
Coin	H						
	T						

b Find the probability of tossing:

i a head and 4

ii a tail and an even number

iii a tail and a number less than 5.

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4 Two dice are rolled and the sum of the numbers is calculated. PS R C

a Copy and complete this table to show all possible sums.

		1st die					
		1	2	3	4	5	6
2nd die	1						
	2					7	
	3						
	4		6				
	5						
	6						

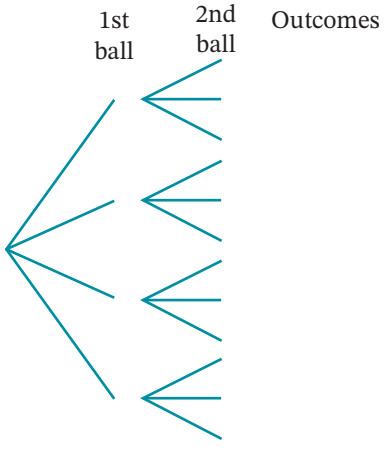
b Find the probability of rolling a sum

i of 3 **ii** of 7 **iii** of at least 6

iv less than 10 **v** that is even **vi** between 3 and 7.

EXAMPLE 10

5 A bag contains 2 blue balls, 1 yellow ball and 1 green ball. 2 balls are drawn out of the bag at random. PS R C



a Copy and complete the tree diagram to list the sample space.

b Find the probability of drawing:

i a blue and green ball, in that order

ii a yellow ball and a green ball, in any order

iii a blue ball and a yellow ball

iv at least one blue ball.

6 The numbers 3, 4, 5 and 8 are written on separate cards and placed in a bag. 2 cards are drawn at random, one after the other, to form a 2-digit number. PS R C

a Use a tree diagram to show all possible outcomes if the cards are drawn *with replacement*, that is, the same card can be used twice.

b Find the probability that the 2-digit number formed:

i ends in 4 **ii** is an odd number

iii is less than 45 **iv** begins with 3 and is even.

7 The weather this weekend will be either fine or rainy each day, with each outcome being equally likely. PS R C

a Draw a tree diagram to show the possible outcomes for Saturday and Sunday.

b What is the probability that:

i it is fine on both days?

ii Saturday is fine and it rains on Sunday?

iii it rains on at least one day of the weekend?

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8 Two cards are chosen at random with replacement from a set of 4 cards consisting of the 2 red Kings and 2 red Queens from a normal deck of cards. Calculate the probability that the cards are: PS R C

a both Kings
b a King and a Queen
c a King and Queen of the same suit
d both Queens
e at least one King
f no Queens

9 Draw a table for the sample space when 2 dice are rolled. Find the probability of rolling: PS R C

a 2 numbers the same
b a 3 and a 6 (in any order)
c at least one 2
d a 4 or a 5 on only one of the dice.

10 While staying at a hotel, Phoebe can choose one item from each course of a breakfast menu. PS R C

1st course	2nd course
Muesli (M)	Bacon and eggs (B)
Raisin toast (R)	Pancakes (P)
Watermelon (W)	Sausages and hash browns (S)
Yoghurt (Y)	

a Find all the different 2-course breakfasts available:
i by listing them
ii using a tree diagram.

b If one of these combinations is chosen at random, find the probability that it includes raisin toast or bacon and eggs.



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11 From a set of cards numbered 1 to 5, 2 cards are drawn at random to make a 2-digit number. PS R C

a Show all outcomes in the sample space:
i by simply listing them
ii using a table.

b What is the probability of making a number:
i that is odd?
ii greater than 30?

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12 Three coins are tossed. P R C

a **i** Use a tree diagram to list the possible outcomes. Show all outcomes in the sample space.

ii How many outcomes are there in the sample space altogether?

b Use your tree diagram to calculate:

i P(only 1 tail) **ii** P(exactly 2 tails)

iii P(no heads) **iv** P(at least 1 head)

Power plus ANSWERS ON P. 605

1 Consider a family with 4 children. Use a tree diagram to list the 16 possible orders of boys and girls in the family. Find the probability of the family having:

a 3 boys and 1 girl **b** no boys

c 2 boys and 2 girls **d** at least 1 girl

e at most one girl **f** more than 2 boys.

2 A bag contains 5 yellow marbles and 4 red marbles. Two draws are made *without replacement*. Calculate the probability of drawing:

a 2 yellow marbles

b a yellow marble first and a red marble second

c a red marble and a yellow marble in any order

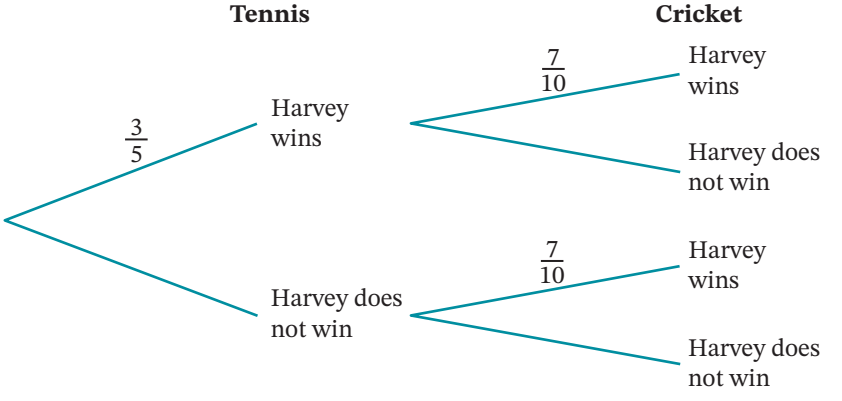
d at most one yellow marble.

3 A more advanced type of tree diagram has probabilities listed on the branches. Harvey is going to play one tennis match and one cricket match. Find the probability that Harvey wins both his tennis and cricket matches.

The probability that he will win the tennis match is $\frac{3}{5}$.

The probability that he and his team will win the cricket match is $\frac{7}{10}$.

This information is shown on some branches of this tree diagram.



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To calculate the probability that Harvey wins both his tennis and cricket matches, we multiply the probabilities along the branches:

$P(\text{Harvey wins tennis and cricket}) = \frac{3}{5} \times \frac{7}{10} = \frac{21}{50}$

Read off the probability that Harvey wins his tennis match and multiply it by the probability that Harvey wins his cricket match.

a If the probability that Harvey wins the tennis match is $\frac{3}{5}$, what is the probability that he does not win the tennis match?

b Hence copy and complete the tree diagram above by writing the missing fractions on the branches.

c Use the tree diagram to calculate:

i $P(\text{Harvey wins only 1 match})$

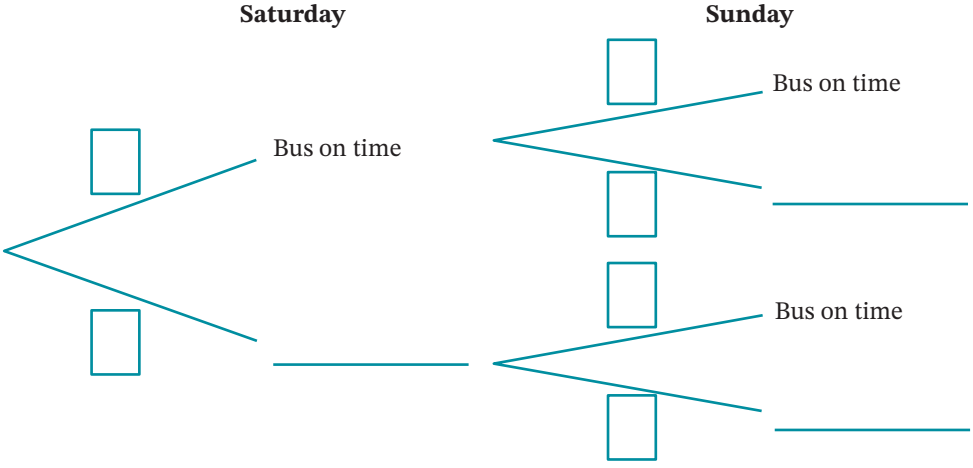
This means Harvey could win one of his matches but not both.

ii $P(\text{Harvey does not win either match})$

iii $P(\text{Harvey wins at least 1 match})$.

4 Eilene takes a bus to her part-time job on Saturdays and Sundays. The probability that the bus will be late on either day is 0.3.

a Copy and complete this tree diagram by writing probability values on the branches and describing the events on the line.



b Calculate the probability that Eilene's bus is

i on time on both days

ii on time on only one of the days

iii late on at least one of the days.

5 Set theory can be used to describe different groups and represent calculations using special notation.

If we consider the set $S = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ that represents all the even numbers from 1 to 20, then $n(S) = 10$, where $n(S)$ represents the *number* of elements in set S.

a Let $E = \{3, 6, 9, 12, 15, 18\}$, which represents the multiples of 3 from 1 to 20.

Find $n(E)$.

b **i** $S \cup E$ represents the **union** of the 2 sets. It consists of all of the elements that are in S or in E or in both S and E.

Copy and complete: $S \cup E = \{ \quad ... \quad \}$

ii $S \cap E$ represents the **intersection** of the 2 sets. It consists of the elements that are common to both S and E.

Copy and complete: $S \cap E = \{ \quad ... \quad \}$

c Given F represents the set of multiples of 5 from 1 to 20, copy and complete:

i $F = \{ ... \}$ **ii** $n(F) =$ ____ **iii** $S \cap F = \{ ... \}$

CHAPTER 12 REVIEW

Language of maths

at least	at most	chance experiment	complementary event
compound event	die/dice	even chance	event
expected frequency	favourable	frequency	mutually exclusive events
observed frequency	observed probability	random	relative frequency
replacement	sample space	theoretical probability	tree diagram
trial	two-stage experiment	two-way table	Venn diagram

Quiz
Language of maths 12

Puzzle
Probability crossword

1 Which term from the above list is used to calculate **observed probability**?

2 What is the meaning of **expected frequency**?

3 What happens to the observed probability of an event as the number of trials becomes very large?

4 Which term describes the set of all possible outcomes of an experiment?

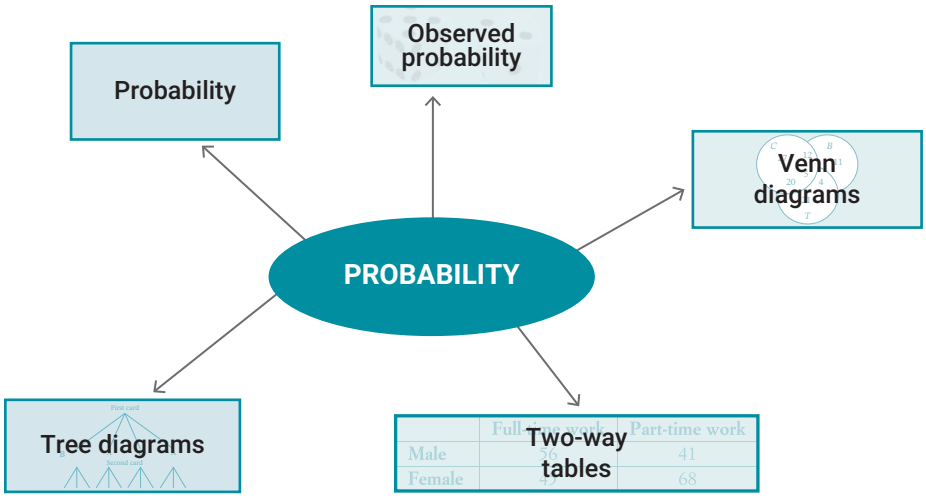
5 Name the event that is **complementary** to winning first prize in Lotto.

6 What is drawn using 2 or more circles, often overlapping?

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Probability



Quiz
Test yourself 12

TEST YOURSELF 12

ANSWERS ON P. 605

12.01

1 A jar contains only red and yellow marbles. The probability of selecting a red marble from the jar is 40%.

a What is the probability of selecting a yellow marble?

b If there are 40 marbles in the bag, how many red marbles are there?

12.01

2 A card is drawn randomly from a normal deck of cards. Find the probability that it is:

a a red card

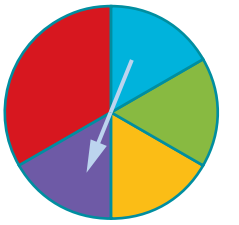
b a King or a Queen card

c a card with a number greater than 6.

12.02

3 For the spinner shown, the red sector is twice as large as each of the other sectors. The spinner was spun 200 times and the results are shown in the table.

Outcome	Frequency
Blue	29
Green	35
Yellow	30
Purple	34
Red	72



a Find the observed probability (as a decimal) that the arrow lands on:

i red **ii** blue or green **iii** not purple **iv** green.

b Find the theoretical probability (as a decimal) that the arrow lands on:

i red **ii** blue or green **iii** not purple **iv** green.

c Are the observed probabilities from part **a** similar to the theoretical probabilities from part **b**?

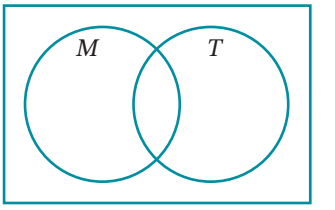
d If the spinner is spun 500 times, calculate the expected frequency of spinning red using the theoretical probability. Select the correct answer **A**, **B**, **C** or **D**.

A 100 **B** 85 **C** 167 **D** 200

PATH

12.03

4 30 teenagers were surveyed on whether they preferred to listen to music (M) or watch TV (T). 21 students preferred music and 18 liked TV, while 4 students did not like either.



a How many students preferred both music and TV?

b Copy and complete the Venn diagram.

c Find the probability of randomly selecting a teenager who likes:

i music but not TV **ii** TV

iii music or TV, but not both **iv** both or neither music nor TV.

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PATH

12.04

5 A group of men and women was surveyed on whether they worked full-time or part-time.

	Full-time work	Part-time work
Male	56	41
Female	45	68

a How many people were surveyed?

b Find the probability (as a percentage, correct to one decimal place) of selecting a person who:

i works full-time

ii is female and works part-time

iii does not work full-time

iv works part-time or is female.

c If a part-time worker is selected at random, what is the probability that the person is female?

12.05

6 4 cards labelled with the letters A, B, C and D are placed in a bag.



2 cards are drawn from the bag at random without replacement.

a List all of the possible outcomes.

b Copy and complete the tree diagram.

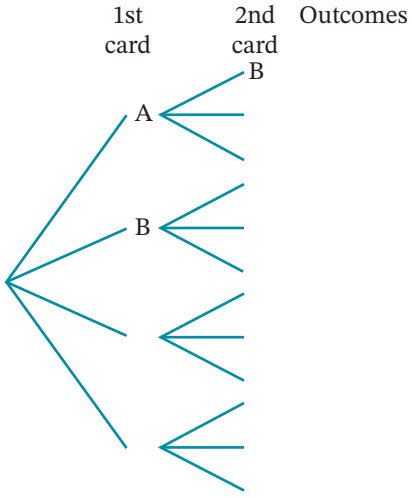
c Use the tree diagram to find the probability of drawing 2 cards with:

i the same letter

ii the letters A and B

iii the letters A or B or both

iv no letter D.



12.06

7 2 dice are rolled and the difference between the numbers is calculated.



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Die 2 \ Die 1	1	2	3	4	5	6
1						
2						
3						3
4						
5				1		
6		4				

6 − 3 = 3

5 − 4 = 1

6 − 2 = 4

a Copy and complete the table to show the sample space.

b Find the probability of rolling a difference of:

i 0 **ii** at least 2 **iii** 4 or 5.

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